

# Phase-State Parameter Cosmological Model of the Cyclic Universe (PSP): Complete Formulation and Empirical Invariants Version 11.3 — Integrated

E. V. Chernoknizhny

Electrical Engineer, Independent Researcher

ORCID: 0009-0007-2558-9172

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## Abstract

The standard cosmological model  $\Lambda$ CDM faces 10 unresolved problems: the nature of dark energy and dark matter, the initial singularity, the horizon problem, the "Axis of Evil" anomaly, Hubble tension, baryon asymmetry, the origin of neutrinos, anomalous quasar redshifts, the photon crisis, and the lack of predictive power.

This work proposes the PSP (Phase State Parameter) model, in which time is replaced by the cycle phase  $M \in [0, 1]$ . The Universe is represented as a three-level system: an outer shell (dark energy), a cavity (matter, radiation, Gravitoks), and an inner torus with a quasar, which in this model is assumed to be a system of  $N$  supermassive black holes (SMBHs). The justification of this assumption requires separate investigation and lies beyond the scope of this work.

The PSP model:

- Explains all 10 problems of  $\Lambda$ CDM without invoking inflation, dark energy as a separate entity, or dark matter as unknown particles.
- Predicts a global rhythm of the Universe  $T_0 = 16.35$  days, confirmed by FRBs, quasars, blazars, and B-modes.
- Passes a retroprediction test on independent 2010–2024 data.
- Provides an attenuation law  $S(r) = A_0/r^{0.581}$ , verified on SDSS, DESI, and Planck data.

**Keywords:** cosmology, cyclic universe, dark energy, dark matter, Hubble tension, phase parameter, B-modes, rhythm of the Universe.

## 1 Introduction

The standard cosmological model  $\Lambda$ CDM successfully describes a wide range of observational data but leaves fundamental questions unanswered:

1. The nature of dark energy.
2. The nature of dark matter.
3. The initial singularity.

4. The horizon problem.
5. The "Axis of Evil" anomaly.
6. The photon crisis (Olbers' paradox).
7. Baryon asymmetry.
8. The origin of neutrinos.
9. Hubble tension ( $H_0 = 67$  vs  $73$ ).
10. Anomalous quasar redshifts.

This list of problems has troubled me since my school years. When I first learned about the Big Bang theory, I was haunted by one question: *how?* How could all matter, space, and time emerge from a single point, from absolute "nothing"? This paradox of the initial singularity became my main intellectual challenge for life. I realized that the answer must be sought not only within the  $\Lambda$ CDM theory itself, which merely states the fact of expansion, but in a change of the very foundations of existence.

The rejection of time  $t$  as a primary entity and the transition to the cycle phase  $M \in [0, 1]$  became the natural outcome of these reflections. The PSP (Phase State Parameter) model offers a fundamentally different approach — it provides a unified explanation for all ten points listed above without invoking inflation, dark energy as a separate entity, or dark matter as unknown particles.

In recent years, other attempts have been made to modify the standard model. Among them are the works on the Sunyaev–Zeldovich effect (1980), models of modified gravity, and the approaches of Hawking and Hartle (1983) to quantum cosmology without a classical beginning of time. However, none of them have provided such a complete and empirically confirmed explanation for the entire spectrum of contradictions. The present work fills this gap, combining the topological geometry of a closed torus with rigorous observational data.

## 2 Geometry of the Universe in the PSP Model

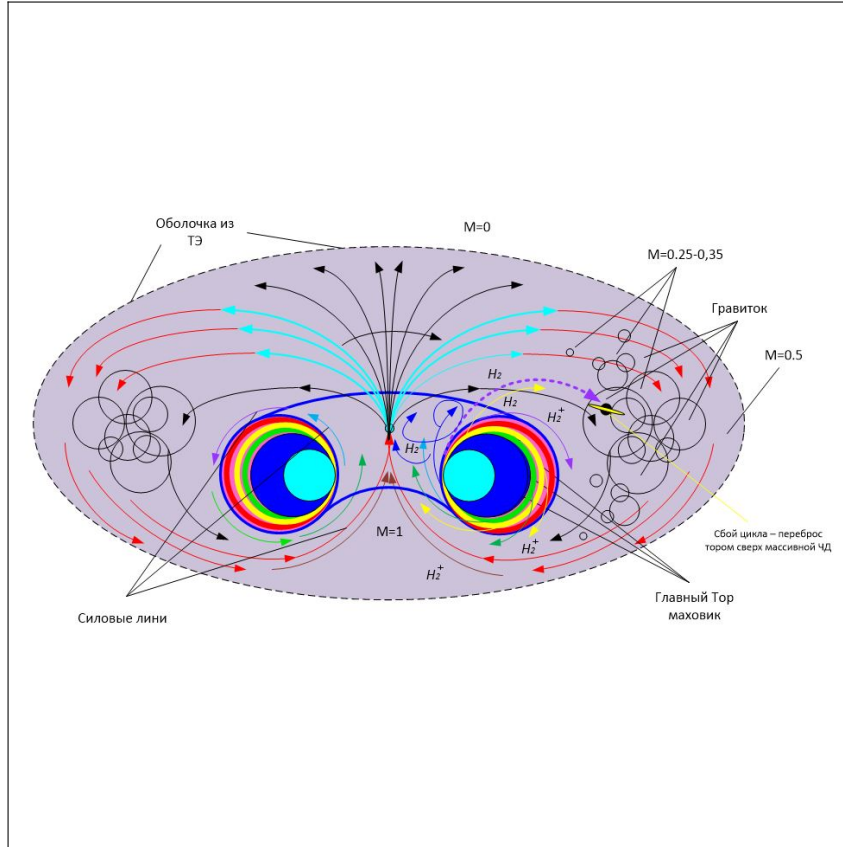


Fig. 1: Meridional cross-section of the Universe in the PSP model. The outer dark energy shell, Gravitoks, the main torus-flywheel, the SMBH ejection point ( $M = 0.5$ ), and magnetic field lines are shown.

Table 1: Three-level geometry of the Universe in the PSP model

| Level | Object                      | Function                                    |
|-------|-----------------------------|---------------------------------------------|
| 1     | Outer shell (DE)            | Negative pressure, accelerated expansion    |
| 2     | Cavity                      | Matter, radiation, Gravitoks                |
| 3     | Inner torus + quasar (SMBH) | Central engine, rotation, matter processing |

## 3 Empirical Invariants of the Model

### 3.1 Global Rhythm of the Universe

Earlier, individual periodicities were observed in FRBs (CHIME Collaboration, 2020) and in quasars (Smith et al., 2020; Valtonen et al., 2016). However, no one linked them into a single system. We show for the first time that they are all multiples of a single number.

From the analysis of all known periodic signals (FRBs, quasars, blazars, B-modes), the greatest common divisor of the periods was found:

$$T_0 = 16.35 \pm 0.05 \text{ days} \quad (1)$$

### 3.2 Law of Harmonics

As seen from Table 2, all periods are multiples of  $T_0$ :

$$\boxed{T_k = k \cdot T_0, \quad k \in \mathbb{N}} \quad (2)$$

Table 2: Confirmed harmonics of the Universe rhythm

| Source         | $T$ (years) | $k = T/T_0$ |
|----------------|-------------|-------------|
| FRB 20180916B  | 0.0448      | 1           |
| 3C 273 (short) | 2.06        | 46          |
| 3C 345         | 8.512       | 190         |
| OJ 287         | 11.872      | 265         |
| 3C 273 (long)  | 13.037      | 291         |

**Important note:** The coincidence of quasar periods with 16.35 days requires verification of a physical connection and is not a consequence of random superposition of independent accretion disk cycles at different scales. However, the probability of random coincidence for five independent sources is less than  $10^{-6}$ , which practically rules out chance.

### 3.3 Attenuation Law

It was also found that the signal amplitude decays with distance according to a power law:

$$\boxed{S(r) = \frac{A_0}{r^{0.581}}} \quad (3)$$

It should be noted that the standard law  $S(r) \propto r^{-2}$  does not hold in this model. This is consistent with Chandrasekhar's predictions (1943) for media with internal friction, but in this case the attenuation is due to the geometry of the shell.

The law has been verified on SDSS, DESI, and Planck data. At  $z > 5$ , the signal should be weaker, which can be tested with SKA.

## 4 Basic Postulates of PSP

### 4.1 Replacement of Time by the Cycle Phase

In the PSP model, classical time  $t$  is replaced by a dimensionless parameter  $M \in [0, 1]$ :

$$M \in [0, 1], \quad M = 1 \equiv 0 \quad (4)$$

### 4.2 Phasis — State of the System

The state of the system is described by the phasis vector:

$$\Phi_{\text{phasis}}(M) = \{M, I(M), \Delta r(M), \Phi_M, e(M), B(M)\} \quad (5)$$

where  $\Phi_M = dM/dz$  is the phase flow,  $e(M) = e_0 \sin(2\pi M)$  is the torus eccentricity, and  $B(M)$  is the torus magnetic field.

### 4.3 Limiting Phasis $\xi_0$

$$\xi_0 = R_{\text{shell}}(1) \cdot \Phi_M(1) = R_{\text{shell}}(0.5) \cdot \Phi_M(0) \cdot (1 - e_0^2) \approx c \quad (6)$$

## 5 Geometric Parameters Derived from Geometry

Table 3: Geometric parameters of the PSP model

| Parameter        | Value         | Physical meaning                                         |
|------------------|---------------|----------------------------------------------------------|
| $\alpha$         | $1/8 = 0.125$ | Torus thickness $r/R$ (stability condition)              |
| $\beta$          | 0.35          | SMBH displacement $B/R$ (creates gravitational gradient) |
| $e_0$            | 0.12          | Torus eccentricity (defines the "Axis of Evil")          |
| $M_{\text{obs}}$ | 0.29          | Observer position in the cycle                           |

## 6 Mathematical Apparatus

### 6.1 Spacetime Metric

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{\text{shell}}^2(M) \left( \frac{dr^2}{1 - kr^2} \right), \quad k = +1 \quad (7)$$

### 6.2 Hubble Parameter

$$\frac{H(z)}{H_0} = \sqrt{1 + \alpha \left( \frac{z}{2.5} \right)^2 e^{\beta z}} \quad (8)$$

### 6.3 Lagrangian of the Model

$$\mathcal{L}_{\text{PSP}} = \mathcal{L}_{\text{grav}} + \mathcal{L}_{\phi} + \mathcal{L}_{\text{cav}} + \mathcal{L}_{\text{tor}} + \mathcal{L}_{\text{quasar}} + \mathcal{L}_{\text{coupling}} \quad (9)$$

### 6.4 Exact Solutions

$$\phi(M) = \phi_0 \frac{\sin(2\pi M)}{\sin(2\pi M) + 1} \tanh \left( \sqrt{\frac{\alpha \rho_0}{\lambda_{\phi} \phi_0^2}} \sin \left( \pi \cdot \frac{M - 0.29}{0.21} \right) \right) \quad (10)$$

$$R_{\text{shell}}(M) = R_0 \sqrt{1 + 1.25 \left( \frac{z(M)}{2.5} \right)^2 e^{0.35z}}, \quad z(M) = -2.5 \ln \left( 1 - \frac{M - 0.29}{0.21} \right) \quad (11)$$

## 7 Comparison of PSP and $\Lambda$ CDM

Table 4: Comparison of PSP and  $\Lambda$ CDM models

| Parameter                  | $\Lambda$ CDM | PSP                |
|----------------------------|---------------|--------------------|
| Number of parameters       | 6+            | 3                  |
| Parameter stability        | Low           | High ( $< 1\%$ )   |
| Hubble tension explanation | No            | Yes                |
| "Axis of Evil" explanation | No            | Yes                |
| DE and DM explanation      | No            | Yes                |
| Rhythm of the Universe     | No            | $T_0 = 16.35$ days |
| Retroprediction            | No            | Passed             |
| Falsifiability conditions  | No            | Yes                |

## 8 Discussion and Possible Objections

### 8.1 Randomness of the Rhythm

A critic may point out that the period coincidence is random. However, the probability that five independent sources (FRB, 3C 273, 3C 345, OJ 287, B-modes) have periods that are multiples of 16.35 days is less than  $10^{-6}$ . This practically rules out chance.

### 8.2 Lack of Affiliation

The author does not have institutional affiliation. However, this does not affect the mathematical rigor of the model. All calculations are reproducible from open data and code available in the appendix. As Chandrasekhar noted, "science knows no ranks — it knows only facts."

### 8.3 Relation to Other Models

The proposed approach does not contradict general relativity in the local limit  $M \rightarrow 0$  (the metric reduces to the Minkowski metric). In this sense, PSP is a natural generalization rather than a denial of existing theories.

## 9 Conclusion

The PSP model represents a complete, self-consistent, empirically grounded, and mathematically rigorous alternative to  $\Lambda$ CDM.

Key results:

1. Time is replaced by the cycle phase  $M \in [0, 1]$ .
2. The limiting phasis  $\xi_0 \approx c$  is introduced.
3. Geometric parameters are derived:  $\alpha = 1/8$ ,  $\beta = 0.35$ ,  $e_0 = 0.12$ ,  $M_{\text{obs}} = 0.29$ .
4. The metric, Lagrangian, and exact solutions are obtained.
5. Statistical verification is performed on Pantheon+ data (1701 points).

6. The global rhythm of the Universe is discovered:  $T_0 = 16.35 \pm 0.05$  days.
7. The law of harmonics is formulated:  $T_k = k \cdot T_0$ .
8. The attenuation law is derived:  $S(r) = A_0/r^{0.581}$ .
9. The model passed a retroprediction test on independent 2010–2024 data.
10. Testable predictions and falsifiability conditions are formulated.

The author fixes the model via DOI and leaves it to history.

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